

1. Preferences & Axioms

Rational preferences: Complete + Transitive.

Completeness: $\forall x, y: x \succsim y$ or $y \succsim x$ (or both).

Transitivity: $x \succsim y, y \succsim z \Rightarrow x \succsim z$.

Strict pref: $x \succ y$ iff $x \succsim y$ and not $y \succsim x$.

Indifference: $x \sim y$ iff $x \succsim y$ and $y \succsim x$.

Monotonicity: More of any good $\Rightarrow$ strictly better. Implies ICs slope down.

LNS: $\forall x, \forall \varepsilon > 0, \exists y$ s.t. $\|y - x\| < \varepsilon$ and $y \succ x$. No thick ICs. Consumer spends full budget.

Convexity: UCS $\{y : y \succsim x\}$ is convex. Mixtures $\geq$ extremes. $\Leftrightarrow$ diminishing [MRS].

Strict convexity: Strict mixtures of distinct equally-good bundles are strictly better. $\Rightarrow$ unique optimum.

Continuity: UCS and LCS are closed. Needed for utility representation.

Utility Representation

Rational + continuous $\Rightarrow \exists$ continuous u representing $\succsim$.

Ordinal: Only ranking matters. If f strictly increasing, $v = f(u)$ represents same $\succsim$. MRS invariant to monotone transforms.

WARP / Revealed Preference

WARP: If x chosen when y affordable, then y never chosen when x affordable and cheaper.

Revealed pref: $x \succsim^* y$ if $\exists B: x, y \in B$ and $x \in C(B)$.

Check WARP: At period-1 prices, is period-1 bundle $\geq$ cost of period-2 bundle? And vice versa at period-2 prices? If both hold with ≥ 1 strict $\Rightarrow$ WARP violated.

IIA (α): Chosen from big set $\Rightarrow$ chosen from any subset.

Sen's β : If x, y both chosen from $B \subseteq A$ and $x \in C(A)$, then $y \in C(A)$.

$\alpha + \beta \Leftrightarrow$ Rationalizable $\Leftrightarrow$ WARP.

2. Utility Functions & MRS

MRS $= \frac{MU_1}{MU_2} = - \frac{dx_2}{dx_1} \Big|_{IC}$

From total diff: $MU_1 dx_1 + MU_2 dx_2 = 0$.

Cobb-Douglas: $u = x_1^\alpha x_2^\beta$

MRS $= \frac{a x_2}{b x_1}$

Equiv: $v = a \ln x_1 + b \ln x_2$.

Smooth, strictly convex ICs. Always interior.

Perfect Substitutes: $u = ax_1 + bx_2$

MRS = a/b (constant). Straight-line ICs.

Perfect Complements: $u = \min\{ax_1, bx_2\}$

L-shaped ICs. Kink at $ax_1 = bx_2$.

MRS: 0 on horizontal, $-\infty$ on vertical, undefined at kink.

Quasilinear: $u = v(x_1) + x_2$

MRS = $v'(x_1)$. Depends only on x_1 . ICs are vertical translates. No income effect for x_1 (at interior).

CES: $u = (x_1^r + x_2^r)^{1/r}$

MRS = $(x_2/x_1)^{1-r}$. Nests: $r \rightarrow 0$ Cobb-Douglas, $r \rightarrow -\infty$ complements, $r = 1$ substitutes.

3. Consumer Choice (UMP)

Budget: $p_1 x_1 + p_2 x_2 \leq m$. Slope = $-p_1/p_2$.

Intercepts: $m/p_1, m/p_2$.

Interior Optimum (Tangency)

$\frac{MU_1}{p_1} = \frac{MU_2}{p_2}$ (bang-per-buck)

Equivalently: MRS = p_1/p_2 .

Lagrangian Method

$\mathcal{L} = u(x_1, x_2) + \lambda[m - p_1 x_1 - p_2 x_2]$

FOCs: $MU_i = \lambda p_i$ for $i = 1, 2$; budget binds.

λ = marginal utility of income.

Cobb-Douglas Demand (Magic Rule)

$u = x_1^a x_2^b$:

$x_1^* = \frac{a}{a+b} \cdot \frac{m}{p_1}, \quad x_2^* = \frac{b}{a+b} \cdot \frac{m}{p_2}$

Spend fraction $\frac{a}{a+b}$ on good 1, $\frac{b}{a+b}$ on good 2.

3 goods: $u = x^a y^b z^c \Rightarrow$ spend $\frac{a}{a+b+c}$ on x , etc.

Perfect Complements Demand

$u = \min\{ax_1, bx_2\}$. Set $ax_1 = bx_2$ + budget:

$x_1^* = \frac{bm}{bp_1 + ap_2}, \quad x_2^* = \frac{am}{bp_1 + ap_2}$

Perfect Substitutes Demand

Compare MU_1/p_1 vs MU_2/p_2 :

- $a/p_1 > b/p_2$: all on good 1: $x_1 = m/p_1$
- $a/p_1 < b/p_2$: all on good 2: $x_2 = m/p_2$
- Equal: any point on budget line

Corner Solutions

If interior FOC gives $x_i < 0$, go to corner: $x_i = 0$, spend all on other good. Common with quasilinear (low income) and perfect subs.

KT: $MU_i - \lambda p_i \leq 0$, equality if $x_i > 0$.

4. Demand Properties & Duality

Marshallian Demand $x_i(p, m)$

Homogeneous degree 0 in (p, m) . Walras' law: $p \cdot x = m$.

Indirect Utility $v(p, m)$

$v(p, m) = u(x^*(p, m))$. Decreasing in p_i , increasing in m .

Roy's Identity: $x_i^* = - \frac{\partial v / \partial p_i}{\partial v / \partial m}$

Expenditure Min (EMP)

$\min p \cdot x$ s.t. $u(x) \geq \bar{u}$.

Hicksian demand $h_i(p, \bar{u})$: demand at prices p to reach utility $\bar{u}$.

Expenditure fn: $e(p, \bar{u}) = p \cdot h(p, \bar{u})$.

Shephard's Lemma: $h_i = \partial e / \partial p_i$.

Duality Relations

$v(p, e(p, \bar{u})) = \bar{u}; \quad e(p, v(p, m)) = m$.

$x_i(p, m) = h_i(p, v(p, m)); \quad h_i(p, \bar{u}) = x_i(p, e(p, \bar{u}))$.

5. Income & Substitution Effects

Slutsky decomposition: Total = Sub + Income.

Price of good 1 rises: **Sub effect** always ≤ 0 (buy less). Income effect depends on normal/inferior.

Compensated bundle: At *new* prices, find bundle on *old* IC (tangent at new price ratio).

Perfect complements: Sub effect = 0. Entire change is income effect. (No reoptimization at kink.)

Normal good: Income effect reinforces sub effect (demand falls when price rises).

Inferior good: Income effect opposes sub effect.

Giffen: Income effect dominates.

6. Choice Under Uncertainty

Expected Utility (vNM)

$EU(L) = \sum p_i u(c_i)$

Axioms: Completeness, Transitivity, Continuity, Independence.

vNM utility unique up to **positive affine** transforms: $\tilde{u} = a + bu, b > 0$. (NOT arbitrary monotone!)

Risk Attitudes

Risk averse $\Leftrightarrow u$ concave $\Leftrightarrow u'' > 0$

$u(E[x]) \geq E[u(x)]$ (Jensen's inequality)

Risk neutral $\Leftrightarrow u$ linear. **Risk loving** $\Leftrightarrow u$ convex.

Certainty Equivalent & Risk Premium

$u(CE) = EU(L) \Rightarrow CE = u^{-1}(EU)$

RP $= E[x] - CE$

Risk averse: $CE < E[x]$, $RP > 0$.

Arrow-Pratt Measures

$r_A(x) = - \frac{u''(x)}{u'(x)}$ (Absolute RA)

$r_R(x) = x \cdot r_A(x) = - \frac{x u''(x)}{u'(x)}$ (Relative RA)

Approx: $RP \approx \frac{1}{2} r_A(\bar{x}) \cdot \sigma^2$.

More risk averse $\Leftrightarrow$ higher r_A everywhere $\Leftrightarrow$ lower CE $\Leftrightarrow$ higher RP $\Leftrightarrow u_2 = g(u_1), g$ concave.

Common Utility Functions

CARA: $u = -e^{-\alpha x}, r_A = \alpha$ (const).

CRRA: $u = \frac{x^{1-\rho}}{1-\rho}$ ($\rho \neq 1$); $u = \ln x$ ($\rho = 1$). $r_R = \rho$.

Log: $r_A = 1/x$ (DARA), $r_R = 1$.

DARA: $r_A' < 0$ – more willing to take fixed-\$ gamble as wealth rises (most realistic).

7. Insurance & Portfolio

Insurance

Wealth: w , loss D w.p. π . Premium q , payout K . $EU = (1 - \pi)u(w - q) + \pi u(w - q - D + K)$

Fair ins ($q = \pi K$): Risk-averse $\Rightarrow$ **full insurance** ($K = D$).

Unfair ($q > \pi K$): partial insurance ($K < D$).

Portfolio Choice

Invest a in risky (return $\tilde{r}$), rest at r_f .

$\bar{W} = w(1 + r_f) + a(\tilde{r} - r_f)$

FOC: $E[u'(\bar{W})(\tilde{r} - r_f)] = 0$

If $E[\tilde{r}] > r_f$: risk-averse invests $a^* > 0$.

Portfolio std dev: $\sigma_P = \alpha \cdot \sigma_{\text{risky}}$ where α = fraction in risky. Expected return: $\alpha E[\tilde{r}] + (1 - \alpha)r_f$.

Contingent Consumption

States $s = 1, \dots, S$ w.p. π_s . $EU = \sum \pi_s u(c_s)$.

$MRS_{12} = \frac{\pi_1 u'(c_1)}{\pi_2 u'(c_2)}$

On 45° line ($c_1 = c_2$): $MRS = \pi_1/\pi_2$.

Fair prices ($q_s = \pi_s$): risk-averse $\Rightarrow c_1 = c_2$ (full insurance).

Stochastic Dominance

FOSD: $F(x) \leq G(x) \forall x \Rightarrow$ all $u' > 0$ prefer F .

SOSD: $\int_{-\infty}^t F(x)dx \leq \int_{-\infty}^t G(x)dx \forall t \Rightarrow$ all risk-averse prefer F . Same mean: G is mean-preserving spread of F .

8. Markets (Gen. Equilibrium)

Edgeworth box: 2 consumers, 2 goods. Dimensions: total endowments.

Feasible: $x_1 + x_2 = E^x, y_1 + y_2 = E^y$.

Pareto efficient: No Pareto improvement exists.

Interior condition: **MRS₁ = MRS₂**.

Contract curve: Set of all PE allocations.

Comp. Equil: Each max utility s.t. budget $p \cdot x_i = p \cdot e_i$, markets clear. Only relative prices matter (normalize $p_y = 1$).

Walras' Law: $p_x \cdot ED_x + p_y \cdot ED_y = 0$. If $n - 1$ markets clear, n -th clears.

Solving: (1) Set $p_y = 1$. (2) Solve each consumer: $MRS_i = p$, budget. (3) Market clearing in one good. (4) Solve for p^* .

1st Welfare Thm: CE $\Rightarrow$ Pareto efficient (needs LNS).

2nd Welfare Thm: Any PE can be CE with transfers (needs convexity).

9. Problem-Solving Techniques

Demand derivation recipe:

1. ID utility type (CD, subs, comp, quasi, CES)
2. CD: magic rule. Comp: set min args equal. Subs: compare bang/buck. Quasi: solve $v'(x_1) = p_1/p_2$, check $x_2 \geq 0$.
3. General: Lagrangian or MRS = p_1/p_2 + budget.
4. Always check $x_i \geq 0$.

EU problem recipe:

1. ID states, probabilities, wealth in each state.
2. Write $EU = \sum p_s u(w_s)$.

- 3. Take FOC w.r.t. choice variable, solve.
- 4. Check corners (e.g., $x = 0$ or $x = w$).

CE calculation: Compute EU , then $u(\text{CE}) = EU$, invert.

WARP check: (1) Cost of bundle A at B's prices vs B's cost. (2) Cost of B at A's prices vs A's cost. (3) Both reveal pref w/ ≥ 1 strict $\Rightarrow$ violation.

Sub/Inc effect: (1) Find original bundle. (2) Find compensated bundle (new prices, old utility). (3) Sub = comp - orig. (4) Inc = new -

comp.

Convexity check: Show $|MRS|$ decreasing along IC (as $x_1 \uparrow$). Or show UCS is convex.

10. Common Exam Traps

- **Marshallian $\neq$ Hicksian.** UMP gives $x(p, m)$; EMP gives $h(p, \bar{u})$. Don't confuse.
- **Perfect comp: sub effect = 0.** Entire change is income effect.

- **WARP inequalities:** "Revealed preferred" = chosen bundle costs $\geq$ alternative *at those prices*.
- **EU states:** Follow probability tree. Final wealth = initial $\pm$ gains/losses.
- **Corners:** If FOC gives $x_i < 0$, set $x_i = 0$.
- "Spend $k\times$ " $\neq$ "Buy $k\times$ " unless prices equal.
- CD exponents must be **normalized** to get spending shares: $\alpha = a/(a + b)$.

- vNM utility: only *affine* transforms ($a + bu$) preserve EU rankings. Not general monotone!
- Portfolio: σ^2 (variance) $\neq \sigma$ (std dev). Square root!
- Quasilinear: interior x_1 demand independent of m , but check $x_2 \geq 0$.
- Risk averse $\Rightarrow$ prefers any mean-preserving contraction.
- \$90 certain vs 50-50 \$0/\$200 is **ambiguous** for risk-averse (lower EV but no risk).